

DBMS — Lecture 2

File System vs DBMS & Advantages of DBMS

Learning Objectives

Understand limitations of traditional file systems.

Compare file systems and DBMS.

Explain major advantages of DBMS.

Traditional File System

Data is stored in separate files designed for individual applications. For example, Student.txt and Fees.txt may both contain Roll No and Name.

Major Problems

Data Redundancy: Same data stored multiple times.

Data Inconsistency: Different files may contain different values for the same data.

Difficult Data Access: New information requirements may require new programs.

Data Isolation: Data is scattered across files and formats.

Security Problems: Fine-grained access control is difficult.

Integrity Problems: Maintaining valid data is harder.

Concurrent Access: Simultaneous updates can conflict.

Backup and Recovery: Recovery after failures can be difficult.

Comparison

Feature	File System	DBMS
Storage	Separate files	Integrated database
Redundancy	Usually high	Can be reduced
Sharing	Limited	Controlled sharing
Security	Limited	Better access control
Concurrency	Difficult	Concurrency control
Recovery	Difficult	Backup and recovery
Data independence	Low	Higher

Advantages of DBMS

Reduced redundancy and improved consistency.

Data sharing among authorized users.

Security and access control.

Integrity constraints.

Transaction and concurrency management.

Backup and recovery.

Data independence.

Key Takeaway

DBMS overcomes many limitations of traditional file-processing systems.